

Integrity Constraints

Reference Integrity Constraints

FOREIGN KEY	-> REFERENCE	DELETE ACTION
PRODUCT .product_type_id	-> PRODUCT_TYPE .product_type_id	ON DELETE RESTRICT
PRODUCT_MARKET .sku	-> PRODUCT .sku	ON DELETE CASCADE
PRODUCT_MARKET .market_id	-> MARKET .market_id	ON DELETE RESTRICT
PRODUCT_MARKET .plug_type_id	-> PLUG_TYPE .plug_type_id	ON DELETE RESTRICT
PRODUCT_MARKET .energy_class_id	-> ENRGY_CLASS .energy_class_id	ON DELETE RESTRICT
COMPONENT .unit_id	-> UNIT .unit_id	ON DELETE RESTRICT
COMPONENT .sub_category_id	-> COMPONENT_SUBCATEGORY .sub_category_id	ON DELETE RESTRICT
COMPONENT_RELATION .parent_stock_number	-> COMPONENT .stock_number	ON DELETE RESTRICT
COMPONENT_RELATION .child_stock_number	-> COMPONENT .stock_number	ON DELETE RESTRICT
COMPONENT_RELATION .unit_id	-> UNIT .unit_id	ON DELETE RESTRICT
COMPONENT_CATEGORY .group_id	-> COMPONENT_GROUP .group_id	ON DELETE RESTRICT
COMPONENT_SUBCATEGORY .category_id	-> COMPONENT_CATEGORY .category_id	ON DELETE RESTRICT
PRODUCT_COMPONENT .sku	-> PRODUCT .sku	ON DELETE CASCADE
PRODUCT_COMPONENT .stock_number	-> COMPONENT .stock_number	ON DELETE RESTRICT
PRODUCT_COMPONENT .unit_id	-> UNIT .unit_id	ON DELETE RESTRICT
PRODUCTION_LINE_PRODUCT .prod_line_id	-> PRODUCTION_LINE .prod_line_id	ON DELETE CASCADE
PRODUCTION_LINE_PRODUCT .sku	-> PRODUCT .sku	ON DELETE CASCADE

FOREIGN KEY	-> REFERENCE	DELETE ACTION
CUSTOMER .market_id	-> MARKET .market_id	ON DELETE RESTRICT
ORDER .customer_id	-> CUSTOMER .customer_id	ON DELETE RESTRICT
ORDER_PRODUCT .order_id	-> ORDER .order_id	ON DELETE CASCADE
ORDER_PRODUCT .sku	-> PRODUCT .sku	ON DELETE RESTRICT
TRANSPORTATION .transportation_type_id	-> TRANSPORTATION_TYPE .transportation_type_id	ON DELETE RESTRICT
TRANSPORTATION .destination_market_id	-> MARKET .market_id	ON DELETE RESTRICT
TRANSPORTATION .order_id	-> ORDER .order_id	ON DELETE CASCADE

Domain Constraints

PRODUCT

- **PRODUCT**.sku -> CHAR(10) NOT NULL, only digits
- **PRODUCT**.volume > 0

COMPONENT

- **COMPONENT**.stock_number -> CHAR(10) NOT NULL, only digits

COMPONENT_RELATION

- **COMPONENT_RELATION**.quantity > 0

PRODUCT_COMPONENT

- **PRODUCT_COMPONENT**.quantity > 0

PRODUCTION_LINE

- **PRODUCTION_LINE**.capacity_par_day > 0

ORDER

- ORDER.status -> ('NEW', 'APPROVED', 'SHIPPED', 'CANCELLED')

ORDER_PRODUCT

- ORDER_PRODUCT.quantity > 0

TRANSPORTATION_TYPE

- TRANSPORTATION_TYPE.capacity > 0

TRANSPORTATION

- TRANSPORTATION.status -> ('NEW', 'APPROVED', 'SHIPPED', 'CANCELLED')

Semantic Integrity Constraints

- Each component belongs to exactly one Group/Category/Subcategory combination
- No infinite loops in component hierarchy, no component can be parent or child component of itself.
- A product can be assigned only to compatible production lines
- An order must contain at least one order item